



B3 Membranes and membrane transport

B · Form and function · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The fluid mosaic model describes the plasma membrane as a phospholipid bilayer with proteins (channel, carrier, receptor) embedded or attached, and cholesterol regulating fluidity; the phospholipids and many proteins can move laterally within the membrane. Passive transport (simple diffusion, facilitated diffusion, osmosis) moves substances down their concentration or water potential gradient without energy input; active transport uses ATP to move substances against their gradient via specific pump proteins. Bulk movement of large materials occurs by endocytosis and exocytosis, involving vesicle formation and fusion with the membrane.

Worked example: follow the reasoning

Potato cylinders of equal mass are placed in sucrose solutions of 0.0, 0.2, 0.4, 0.6, 0.8 and 1.0 mol dm⁻³ and percentage change in mass is measured after one hour, giving a straight-line trend from +6% at 0.0 mol dm⁻³ to -10% at 1.0 mol dm⁻³. Step 1: plot % mass change (y) against sucrose concentration (x). Step 2: find the x-intercept, i.e. the concentration where % mass change = 0 — here it occurs at approximately 0.6 mol dm⁻³ (reading from the linear trend). Step 3: at this concentration, the external solution has the same water potential as the potato cells (isotonic), so there is no net water movement. Conclusion: the sucrose concentration of the potato cell cytoplasm is approximately 0.6 mol dm⁻³.

Quick check · multiple choice

Which of the following transport processes requires ATP to move a solute against its concentration gradient?

- A Facilitated diffusion
- B Osmosis
- C Active transport
- D Simple diffusion through the phospholipid bilayer

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A student places identical potato cylinders into distilled water and into a concentrated sucrose solution for 30 minutes, then measures the change in mass of each cylinder. [6]
 - (a) Predict, with reasons, the change in mass of the potato cylinder placed in distilled water. [2]
 - (b) Predict, with reasons, the change in mass of the potato cylinder placed in the concentrated sucrose solution. [2]
 - (c) Explain, in terms of water potential, why water moves into or out of the potato cells in each case. [2]

Final answer: _____



Answers and explanation

Question 1 [6]

- (a) The cylinder in distilled water gains mass, because distilled water has a higher (less negative) water potential than the potato cell cytoplasm, so water moves into the cells by osmosis, increasing their mass (cells become turgid, limited by the cell wall).
- (b) The cylinder in the concentrated sucrose solution loses mass, because the external solution has a lower (more negative) water potential than the cell cytoplasm, so water moves out of the cells by osmosis, decreasing their mass (cells become flaccid/plasmolysed).
- (c) Water moves by osmosis from a region of higher (less negative) water potential to a region of lower (more negative) water potential across the partially permeable plasma membrane; in distilled water the external solution has higher water potential than the cytoplasm so water enters, while in concentrated sucrose the external solution has lower water potential than the cytoplasm so water leaves.

Quick check answer

C. Active transport uses energy from ATP hydrolysis to power a carrier/pump protein, allowing net movement of a solute against its concentration gradient.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.